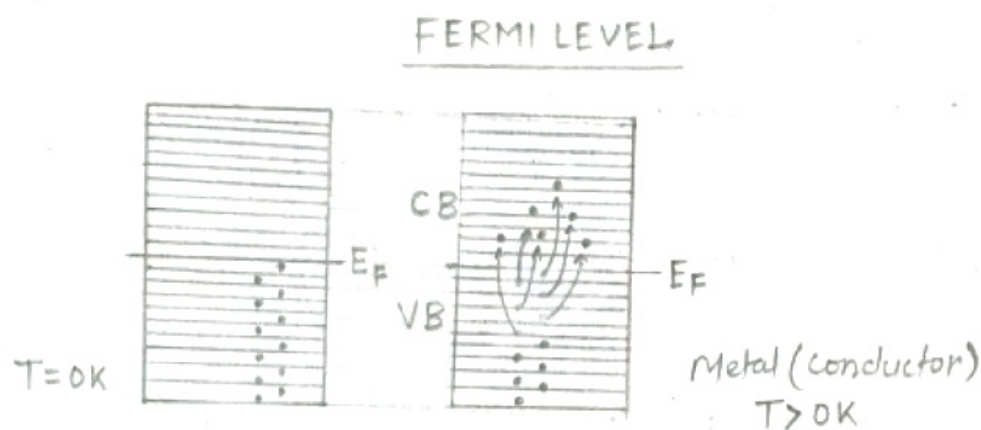


Fermi-Dirac Probability function

Fermi level :-

- In Metal, there is one partially filled band which is a result of conduction band overlapping with valence band.
- In this band lowest energy levels are filled first.
- The highest occupied energy level at absolute zero temp (0K) is called Fermi level.
- Energy corresponding to it is called as Fermi energy denoted by E_F .



At temp $T > 0K$, the distribution of electrons over a range of allowed energy levels at thermal equilibrium is given by Fermi-Dirac distribution function.

$$f(E) = \frac{1}{1 + e^{(E - E_F)/kT}}$$

where, $f(E) \rightarrow$ Probability of occupancy for energy level E .

$E_F \rightarrow$ Fermi Energy

$T \rightarrow$ Temp in 0K (zero calvin)

(Boltzmann const) $k = \frac{1.38 \times 10^{-23} \text{ J/K}}{1.6 \times 10^{19}} = 8.437 \times 10^{-5} \text{ eV/K}$

Fermi Dirac function

$$kT \rightarrow V_T \text{ (Thermal voltage)}$$

$$f(E) = \frac{1}{1 + e^{(E - E_F)/kT}}$$

Case 1:- $T = 0K$

$$E > E_F \quad f(E) = \frac{1}{1 + e^{+\infty}} = 0$$

{ This indicates all energy levels above E_F are completely empty at absolute zero temperature. }

$$E < E_F \quad f(E) = \frac{1}{1 + e^{-\infty}} = 1$$

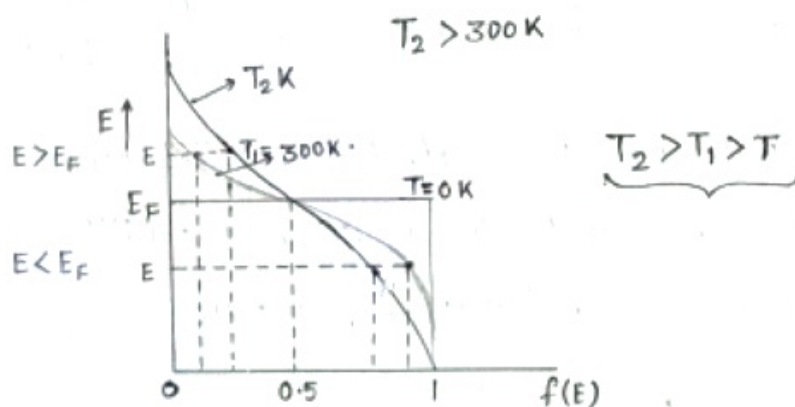
{ This indicates all energy levels below E_F are completely filled at absolute zero temperature. }

Case 2:- $T \neq 0K$

$$E = E_F \quad f(E_F) = \frac{1}{1 + e^{0/kT}} = 0.5 \quad [e^{0/kT} = 1]$$

{ At $T > 0K$, probability of occupancy for Fermi level is always half }

Fermi Dirac distribution curve



$$E > E_F$$

$$P_2 > P_1 > P_0$$

$P \rightarrow$ Probability

$$E < E_F$$

$$P_2 < P_1 < P_0$$